

A SELF-DUAL POLYTOPE WITHOUT A SELF-POLAR REALIZATION

MARTIN WINTER[†]

ABSTRACT. We construct a realizable, combinatorially self-dual 10-dimensional polytope with 173 vertices and 173 facets whose combinatorial type admits no self-polar realization. This answers a question of Jensen on the existence of self-polar realizations in the negative. The construction retains a 4-dimensional face of Bokowski–Ewald–Kleinschmidt type and forces every combinatorial duality to square to its nonrealizable involution on that face. Two edge truncations and two dual ridge additions make the required markings intrinsic. An elementary argument with homogenized cones converts the resulting projective symmetry into an affine symmetry, contradicting the Bokowski–Ewald–Kleinschmidt obstruction.

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For more information, see: martinwintermath.github.io/ai-math.html

1. INTRODUCTION

Polarity realizes the dual face lattice of a convex polytope. If that lattice is self-dual, it is natural to ask whether the realization can be chosen so that polarity is also a geometric symmetry. For a full-dimensional polytope $P \subset \mathbb{R}^d$ with $0 \in \text{int } P$, write

$$P^\circ = \{y \in \mathbb{R}^d \mid \langle y, x \rangle \leq 1 \text{ for all } x \in P\}.$$

Following Jensen [3], we call P *self-polar* if $P = UP^\circ$ for some orthogonal transformation U . We reserve *self-dual* for the combinatorial condition that the face lattice is isomorphic to its dual. The question is whether self-duality guarantees a self-polar realization.

For a polytope type, admitting a self-polar realization is equivalent to admitting a realization isometric, linearly equivalent, affinely equivalent, or projectively equivalent to its polar. More precisely, every realization projectively equivalent to its polar has a projectively equivalent self-polar realization. This normalization

[†]MAX-PLANCK INSTITUTE FOR MATHEMATICS IN THE SCIENCES, INSELSTRASSE 22, 04103 LEIPZIG, GERMANY

Email address: martin.winter@mis.mpg.de.

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is proved in [Proposition B.1](#) in [Appendix B](#). The equivalence concerns the existence of realizations; the conditions need not coincide for a fixed realization.

There is good reason to expect a positive answer in small dimensions. Regular polygons admit self-polar realizations, and Jensen proved that every self-dual 3-dimensional polytope admits one [\[3\]](#). The 3-dimensional argument uses the uniqueness of a normalized realization with edges tangent to a sphere. Jensen then asked [\[3\]](#):

Does every self-dual polytope have a self-polar realization?

Equivalent formulations using linear, affine, and projective transformations were also raised in [\[5\]](#).

We answer this realization question in the negative.

Theorem 1.1. *There exists a realizable, combinatorially self-dual 10-dimensional polytope type with 173 vertices and 173 facets that admits no self-polar realization.*

To our knowledge, this is the first example of a realizable self-dual polytope type with no self-polar realization. We do not claim that its dimension or its number of vertices is minimal.

The obstruction comes from a 4-dimensional polytope of Bokowski, Ewald, and Kleinschmidt [\[1\]](#). It has a combinatorial involution that cannot be induced by an affine automorphism in any realization. We retain a face of this type and force *every* duality of the ambient polytope to square to that involution on the face. An elementary argument with homogenized cones shows that a projective realization of this involution would yield an affine realization. This is where the geometric contradiction occurs.

2. A HIGH-LEVEL EXPLANATION

The construction has three stages. We describe their roles before introducing the face labels.

First, let P be the Bokowski–Ewald–Kleinschmidt polytope, with its nonrealizable involution α . The join $Q = P \star P^*$ is automatically self-dual: a duality can exchange the two factors. We instead use a twisted exchange, applying α to one of the factors. Its square acts as α on both factors. The difficulty is that the ordinary, untwisted exchange remains available. Exhibiting one forbidden duality does not exclude a self-polar realization inducing a different duality.

Second, we place a prism over Q beneath a pyramid over its upper base. This *roofed prism* has a bottom copy and a middle copy of the vertices of Q , together with an apex. Its duality exchanges bottom vertices with roof facets and middle vertices with side facets. We then truncate two bottom edges and perform the dual operations at two roof ridges. The four selected faces form a four-cycle under the twisted duality, so the final polytope remains self-dual.

These operations also record two kinds of marking. The cut facets mark two pairs of vertices of Q ; the added roof vertices mark two pairs of facets of Q . Incidence counts and graph neighborhoods recover these markings without using the construction labels. Their compatibility forces every automorphism on the middle boundary to be either the identity or the simultaneous action of α . Consequently, every duality has the same square there.

Finally, the middle copy of the P factor survives as an actual 4-dimensional face. A self-polar realization would induce a duality whose square is a projective automorphism inducing α on this face. After changing

the affine section of its homogenized cone, this action becomes affine, contrary to the defining obstruction of P .

The dimension count is now visible: the join has dimension nine, since $4 + 4 + 1 = 9$, and the roofed prism adds one dimension. The remaining work is to justify the local operations and make the two markings intrinsic.

3. THE INPUT POLYTOPE AND THE TWISTED JOIN

We include the input data to fix the labeling of the external obstruction. Let P be the simplicial 4-dimensional polytope on vertices $0, \dots, 9$ with the 28 tetrahedral facets in [Table 1](#). Here a string such as 0126 denotes the vertex set $\{0, 1, 2, 6\}$.

0126	0137	1278	1689
0236	1237	2789	1269
0356	0457	5789	4568
3456	0347	4579	4689
0156	0157	1578	1568
2346	2347	2479	2469
0345	0123	4589	1289

Table 1. The facets of the Bokowski–Ewald–Kleinschmidt polytope in the labeling used throughout.

Set

$$\alpha = (0\ 9)(1\ 4)(2\ 5)(3\ 8)(6\ 7).$$

Theorem 3.1 (Bokowski–Ewald–Kleinschmidt). *The type P is realizable, and α is a combinatorial automorphism of P . In no realization of this labeled type is α induced by an affine automorphism.*

The obstruction for precisely this involution is stated in the final remark of Section 2 of [1, p. 125]. The automorphism used earlier in that paper’s main argument is different. After subtracting one from each vertex label, its Table 1 agrees with [Table 1](#); realizability is established in its Sections 3–4. With the same relabeling, the labeling and scope of [Theorem 3.1](#) are also recorded in [2, Section 3.3 and Table 1]. This is the only non-elementary external input in the proof of [Theorem 1.1](#).

We write $\mathcal{L}(P)$ for the full face lattice of a polytope P , and $\mathcal{V}(P)$ and $\mathcal{H}(P)$ for its vertices and facets. A duality is an order-reversing bijection of the full face lattice, including its empty and full faces. Composition acts from right to left. The symbol $*$ denotes abstract face duality, while $\star$ denotes the join.

We shall use the marked incident pair

$$v = 0, \quad F = \{0, 1, 2, 6\}.$$

Thus $v \in F$, but $v \notin \alpha(F)$ and $\alpha(v) \notin F$.

Lemma 3.2. *The stabilizer of (v, F) in $\text{Aut}(P)$ is trivial. Moreover, P and P^* are nonisomorphic and join-indecomposable.*

Proof. The numbers of tetrahedral facets containing the edges 01, 02, and 06 are 5, 3, and 4, respectively. An automorphism fixing v and F must therefore fix all four vertices of F . It fixes a full flag in F , extended by P . A polytope automorphism fixing a full flag is the identity: the adjacent flag of any given rank is unique, so flag fixing propagates across the connected flag graph.

The two types have 10 and 28 vertices, so they are nonisomorphic. The facet list shows that no vertex of P belongs to all but one facet, hence P is not a pyramid. A simplicial polytope that is a nontrivial join is either a pyramid or a simplex. Indeed, if both factors have positive dimension, each occurs as a face of a simplicial facet of the join and must be a simplex; their join is then a simplex. This excludes a join decomposition of P . Dualizing a join decomposition gives a join decomposition of the dual face lattice, so P^* is also join-indecomposable. $\square$

Form $Q = P \star P^*$. It is a realizable 9-dimensional polytope with 38 vertices and 38 facets. Every face is uniquely $A \star B^*$ with $A, B \in \mathcal{L}(P)$. Define

$$\delta_Q(A \star B^*) = \alpha(B) \star A^*.$$

This is a duality, and

$$(3.1) \quad \delta_Q^2 = \sigma, \quad \sigma(A \star B^*) = \alpha(A) \star \alpha(B)^*, \quad \sigma^2 = \text{id}.$$

Thus the square remembers the obstruction on each factor.

For $u \in \mathcal{V}(P)$ and $G \in \mathcal{H}(P)$, name the facets of Q by

$$A_u = P \star u^*, \quad B_G = G \star P^*.$$

The vertices of Q are u and G^* . Directly from the definition,

$$\begin{aligned} \delta_Q(u) &= A_u, & \delta_Q(G^*) &= B_{\alpha(G)}, \\ \delta_Q(A_u) &= \alpha(u), & \delta_Q(B_G) &= G^*. \end{aligned}$$

The join also admits the untwisted factor exchange. The local modifications below are needed to rule out that alternative.

4. THE ROOFED PRISM AND ITS FOUR MODIFICATIONS

Realize Q with the origin in its interior. Take the prism $Q \times [0, 1]$ and add the point $a = (0, 1 + \varepsilon)$ beyond its upper base. This point is strictly beneath the bottom and every vertical facet. The resulting 10-dimensional polytope R is a roofed prism, a construction described in [4, Section 3.2].

Write its bottom and middle vertices as b_x and m_x for $x \in \mathcal{V}(Q)$. Its facets are

$$\begin{aligned} D &= \text{conv}\{b_x \mid x \in \mathcal{V}(Q)\}, \\ S_H &= \text{conv}\{b_x, m_x \mid x \in \mathcal{V}(H)\}, \\ T_H &= \text{conv}(\{a\} \cup \{m_x \mid x \in \mathcal{V}(H)\}) \quad (H \in \mathcal{H}(Q)). \end{aligned}$$

In particular, R has 77 vertices and 77 facets.

The twisted duality extends to a duality δ_R by

$$\begin{aligned}\delta_R(b_x) &= T_{\delta_Q(x)}, & \delta_R(m_x) &= S_{\delta_Q(x)}, \\ \delta_R(T_H) &= b_{\delta_Q(H)}, & \delta_R(S_H) &= m_{\delta_Q(H)},\end{aligned}$$

and $\delta_R(a) = D$, $\delta_R(D) = a$. To verify the extension it suffices to check vertex–facet incidences. Each nontrivial incidence reduces to $x \in H$ in Q and its equivalent dual incidence $\delta_Q(H) \in \delta_Q(x)$.

Choose the bottom edges and roof ridges

$$\begin{aligned}e_0 &= [b_v, b_{F^*}], & e_1 &= [b_{\alpha(v)}, b_{\alpha(F)^*}], \\ h_0 &= T_{A_v} \cap T_{B_{\alpha(F)}}, & h_1 &= T_{A_{\alpha(v)}} \cap T_{B_F}.\end{aligned}$$

Every cross-factor vertex pair of a join is an edge, so the edges exist; their dual faces are ridges. They form the cycle

$$(4.1) \quad e_0 \xrightarrow{\delta_R} h_0 \xrightarrow{\delta_R} e_1 \xrightarrow{\delta_R} h_1 \xrightarrow{\delta_R} e_0.$$

We define S by shallowly truncating e_0, e_1 and adding vertices z_0, z_1 beyond exactly the two facets containing h_0, h_1 , respectively. Denote the new cut facets by C_0, C_1 . The following proposition makes this prescription unambiguous.

Proposition 4.1. *The four operations can be realized convexly in any order. Their resulting face lattice is independent of order under the natural identification of the new vertices and facets. The polytope S is self-dual, and it has a duality δ_S satisfying*

$$(4.2) \quad \delta_S(m_x) = S_{\delta_Q(x)}, \quad \delta_S(S_H) = m_{\delta_Q(H)}.$$

Proof. We first describe each operation geometrically. To truncate an edge e , choose a supporting affine functional with maximizer set exactly e and cut by a sufficiently close parallel hyperplane. This removes its two endpoints and no other vertex. There is one new vertex on each edge having exactly one endpoint in e . A new vertex belongs to an old facet exactly when that facet contained both endpoints of its crossing edge. Every old facet retains relative interior points when the cut is sufficiently shallow, and therefore remains a facet.

For a ridge addition, let p be in the relative interior of the ridge. Write the facet inequalities as $\langle n_H, x \rangle \leq b_H$. Only the two containing facet inequalities are tight at p . Their normals are independent, so we can choose a direction w with positive scalar product with both. For sufficiently small $t > 0$, the point $p + tw$ violates those two inequalities and satisfies every other one strictly.

The visible facets are therefore exactly the prescribed pair. They are replaced by cones from the new vertex over their horizon ridges, namely the ridges shared by a visible and a nonvisible facet. All old vertices survive: each belongs to at least ten facets, and a retained incident facet has its original vertex set and geometry.

The two bottom edges have disjoint endpoint sets. Choose the cut depths so small that neither cut removes vertices introduced by the other. If an old edge connects the two endpoint sets, its middle segment survives and carries one cut vertex at each end. The incidences of those vertices depend only on the original crossing edge and the relevant cut facet. This identifies the two orders canonically.

For the roof additions, disjoint visible facet pairs do not imply disjoint boundaries. We therefore consider their possible interaction. Neither target ridge lies in any facet removed by the other operation: each ridge lies in exactly its two prescribed facets. Both containing facets are retained by the other addition, with their original intersection. Thus the second target remains a ridge. Every newly introduced facet is strict at its relative interior, and a sufficiently close second added point sees exactly its intended pair.

It remains to track the horizon. Suppose a ridge r originally separates a facet visible to the first addition from one visible to the second. After the first addition, r is the intersection of the retained original facet with $\text{conv}(z_0, r)$. The second addition retains this cone and creates $\text{conv}(z_1, r)$. Reversing the order produces the same two cones. Every other horizon ridge retains its vertex set and produces its single prescribed cone. Hence the roof additions commute combinatorially; no facet containing both new vertices occurs.

Finally, the bottom cuts change no roof face, including its middle bases. Roof additions introduce no bottom vertices. The two kinds of operation therefore commute as well. These statements concern the combinatorial type; the geometric realizations need not coincide.

Edge truncation is dual to adding a vertex beyond the ridge corresponding to the edge: its two endpoints correspond to the two visible facets. The cycle (4.1) permutes the complete set of modifications. Their order independence therefore induces a duality of the final face lattice. The middle vertices and side facets survive throughout, giving (4.2). $\square$

Proposition 4.2. *The faces of S whose vertex sets are contained in $M = \{m_x \mid x \in \mathcal{V}(Q)\}$ form precisely the middle copy of the boundary of Q . In particular,*

$$P_M = \text{conv}\{m_u \mid u \in \mathcal{V}(P)\}$$

is a genuine 4-dimensional face of type P .

Proof. For every facet H of Q , the middle copy m_H survives. If T_H is retained, it is $S_H \cap T_H$. If T_H is removed, its middle base $S_H \cap T_H$ is a horizon ridge and gives a new facet $\text{conv}(z_i, m_H)$. Intersecting that facet with S_H again gives m_H . The later operations preserve these faces.

Intersections of the middle facet copies recover every proper face of Q . Conversely, the trace on M of every final facet is a proper face of Q or empty: it is a middle facet for a side or roof facet, a middle face for an added roof facet, and empty for the bottom or a cut facet. Since every face is an intersection of facets, there are no additional faces on M .

In particular, intersect the middle faces m_{A_u} over all $u \in \mathcal{V}(P)$. Their intersection is precisely the middle P factor, because the intersection of all facets u^* of P^* is empty. Hence P_M is a face with the required lattice and dimension. $\square$

The set M itself is not a face. Only its induced boundary is used, together with the genuine face P_M . These are statements about the face lattice, so they hold in every realization of S .

5. RECOVERING THE MARKINGS

We next show that the modifications can be recognized without their labels. The only numerical ingredient is the finite incidence count in Table 2. It follows from Table 1 and the local rules in the proof of Proposition 4.1; explicit enumeration procedures are given in the appendix.

Each selected bottom endpoint pair has graph degrees 36 and 15 in R . For example, v has seven neighbors in P , all 28 vertices of the other join factor, and its vertical partner. The vertex F^* has four neighbors in P^* , all ten vertices of P , and its vertical partner. Each cut therefore creates

$$(36 - 1) + (15 - 1) = 49$$

vertices. Each roof addition has 49 horizon ridges (by the dual edge count) and creates 49 facets. The independence of the modifications gives

$$f_0(S) = 77 - 4 + 2 \cdot 49 + 2 = 173,$$

$$f_9(S) = 77 + 2 - 4 + 2 \cdot 49 = 173.$$

Vertex class	Incident-facet count, with multiplicity
a	128 (1)
z_0, z_1	49 (2)
Middle vertices from P	74 (2), 85 (2), 93 (6)
Middle vertices from P^*	80 (8), 109 (20)
Remaining bottom vertices	21 (2), 23 (6), 33 (26)
Cut vertices	15 (4), 16 (64), 17 (4), 18 (14), 21 (2), 33 (10)

Table 2. Exact incidence counts obtained by finite enumeration from the input facet list. The parenthesized number is the multiplicity.

For illustration, a belongs to the original 38 roof facets, loses four of them, and belongs to 47 new facets from each roof addition. The other two new facets are the cones over the middle bases. Hence its count is $38 - 4 + 2 \cdot 47 = 128$.

Lemma 5.1. *Every automorphism of S fixes a and preserves the unordered pairs $\{z_0, z_1\}$ and $\{C_0, C_1\}$. It also preserves M , its middle boundary, and the family of side facets.*

Proof. The histogram distinguishes a and the pair of roof vertices. By self-duality, the facet-size histogram is the same; thus C_0, C_1 are exactly the facets with 49 vertices.

Let $N_S(w)$ denote the graph neighborhood of a vertex w . The local incidence rules give

$$(5.1) \quad M = N_S(a) \setminus \{z_0, z_1\}.$$

Indeed, the old neighbors of a are precisely the middle vertices; they survive the operations, and its only new neighbors are z_0, z_1 . This preserves the middle boundary by [Proposition 4.2](#).

The side facets are exactly the facets avoiding a, z_0, z_1 and meeting M . The bottom and cut facets do not meet M ; retained roof facets contain a ; and new roof facets contain an added vertex. This characterizes the side facets intrinsically. $\square$

There is one additional graph observation. Each m_x has exactly one neighbor outside $M \cup \{a, z_0, z_1\}$. If b_x survives, it is b_x . If b_x is removed, it is the cut vertex on the original vertical edge $b_x m_x$. This follows either directly from the crossing-edge rule or from the facet-closure graph test in the appendix.

Call this neighbor w_x . Then $w_x \in C_i$ if and only if x is an endpoint of the marked edge underlying e_i . Thus the cuts recover the unordered marking family

$$(5.2) \quad \{\{v, F^*\}, \{\alpha(v), \alpha(F)^*\}\}$$

on Q . The use of w_x is essential: the middle vertex itself belongs to neither cut facet.

Exactly two facets through z_i avoid a . They are the cones over its two visible middle bases. Their traces on M recover the unordered family

$$(5.3) \quad \{\{A_v, B_{\alpha(F)}\}, \{A_{\alpha(v)}, B_F\}\}.$$

These observations show that an automorphism induces an automorphism γ of Q preserving both marking families. Moreover, side-facet recognition gives $S_H \mapsto S_{\gamma(H)}$, because each side facet has its unique middle facet trace.

6. THE SQUARE OF EVERY DUALITY

We now determine the possible restrictions to the middle boundary. Recall that the connected components of the bipartite vertex–facet *nonincidence* graph recover the indecomposable join factors. A join has no cross-factor nonincidences. Conversely, a block decomposition of nonincidences makes face closures split into the corresponding factors, giving the join decomposition. This also shows that duality preserves join-indecomposability.

Since P and P^* are indecomposable and nonisomorphic, any automorphism of Q preserves the two factors. Identifying $\text{Aut}(P^*)$ with $\text{Aut}(P)$ through abstract duality, write $\gamma = (g, h) \in \text{Aut}(P) \times \text{Aut}(P)$. Its action on the named facets is

$$\gamma(A_u) = A_{h(u)}, \quad \gamma(B_G) = B_{g(G)}.$$

Proposition 6.1. *Every automorphism of S induces either id or σ on the middle copy of Q .*

Proof. Let $\varepsilon \in \{0, 1\}$ indicate whether the automorphism exchanges C_0, C_1 , and let $\eta \in \{0, 1\}$ indicate whether it exchanges z_0, z_1 . The two factors are preserved, so (5.2) gives

$$g(v) = \alpha^\varepsilon(v), \quad h(F) = \alpha^\varepsilon(F).$$

Using *both* facet pairs in (5.3) gives

$$g(F) = \alpha^\eta(F), \quad h(v) = \alpha^\eta(v).$$

For example, when the roof pairs are preserved, the second pair fixes $g(F)$ and the first fixes $h(v)$; when they are exchanged, these images are $\alpha(F)$ and $\alpha(v)$. No commutation of g or h with α is assumed.

Because g preserves the incidence $v \in F$, unequal values of ε, η would imply $v \in \alpha(F)$ or $\alpha(v) \in F$. Both are false. Hence $\varepsilon = \eta$. The maps $\alpha^{-\varepsilon}g$ and $\alpha^{-\varepsilon}h$ now fix (v, F) . By Lemma 3.2, they are the identity. Thus $g = h = \alpha^\varepsilon$, as required. $\square$

Only this restriction is needed. An automorphism in its kernel would not affect the following argument.

Proposition 6.2. *Every duality Δ of S satisfies*

$$(6.1) \quad \Delta^2(m_x) = m_{\sigma(x)} \quad (x \in \mathcal{V}(Q)).$$

In particular, Δ^2 preserves P_M and induces α there.

Proof. The map $t = \Delta\delta_S^{-1}$ is an automorphism, so $\Delta = t\delta_S$. By [Proposition 6.1](#), its restriction to Q is $\gamma \in \{\text{id}, \sigma\}$. Side-facet recognition and [\(4.2\)](#) yield

$$\Delta(m_x) = S_{\gamma\delta_Q(x)}, \quad \Delta(S_H) = m_{\gamma\delta_Q(H)}.$$

Consequently,

$$\Delta^2(m_x) = m_{\gamma\delta_Q\gamma\delta_Q(x)}.$$

By [\(3.1\)](#), $\sigma = \delta_Q^2$ commutes with δ_Q and has square the identity. Either choice of γ therefore gives [\(6.1\)](#). Its restriction to the middle P factor is precisely α . $\square$

This is the central combinatorial conclusion. Order four of a particular duality would not suffice: the obstruction must apply to every duality induced by a self-polar realization.

7. FROM PROJECTIVE SYMMETRY TO AFFINE SYMMETRY

We have forced a nonrealizable affine action on a face. We use the projective formulation of self-polar realizability from [Proposition B.1](#). The following elementary lemma excludes a projective realization of the prescribed action on that face.

Lemma 7.1. *Let P be a join-indecomposable polytope type, and let $\beta \in \text{Aut}(P)$ be an involution. If β is induced by a projective automorphism in some realization of P , then it is induced by an affine automorphism in some realization of P .*

Proof. Take the homogenized cone

$$C = \text{cone}\{(x, 1) \mid x \in P\}$$

over such a realization. It is pointed and full-dimensional. A lift of the projective transformation has an affine denominator that is nonzero throughout P , because its image lies in the same bounded affine chart. The denominator has constant sign on the connected set P . Choosing the overall sign of the lift therefore gives an invertible linear map T with $T(C) = C$, mapping each extreme ray to a positive multiple of its prescribed image ray.

Since $\beta^2 = \text{id}$, the map T^2 fixes every extreme ray individually. The cone is indecomposable because a nontrivial direct sum would make its bounded section a nontrivial join. By [Lemma B.3](#), T^2 is a positive scalar matrix.

Hence $T^2 = \lambda I$ for some $\lambda > 0$. Replace T by $\lambda^{-1/2}T$ so that $T^2 = I$, without changing its projective action or its action on the cone. Choose a covector ℓ strictly positive on $C \setminus \{0\}$. Both ℓ and $T^\top \ell$ are strictly positive there, so

$$\bar{\ell} = \frac{1}{2}(\ell + T^\top \ell)$$

is strictly positive and $T^\top$ -invariant. The section $\{x \in C \mid \bar{\ell}(x) = 1\}$ is a bounded realization of the same type, and T restricts to an affine automorphism inducing β . $\square$

The original projective transformation need not have been assumed to have order two as a geometric map. Finite order of its action on the extreme rays is enough for the argument.

Proof of Theorem 1.1. By Proposition 4.1, S is convexly realizable, 10-dimensional, and self-dual. The vertex and facet counts were established above.

Suppose some realization of S is projectively equivalent to its polar. By Lemma B.2, it induces a duality Δ whose square is realized by a projective automorphism. By Proposition 6.2, this automorphism preserves the genuine face P_M and induces α there. Restrict its cone lift to the 5-dimensional linear span of the cone over P_M . This gives a projective realization of α on the BEK type.

That type is join-indecomposable by Lemma 3.2. Hence Lemma 7.1 gives an affine realization of α , contradicting Theorem 3.1.

This excludes every realization projectively equivalent to its polar and hence every self-polar realization by Proposition B.1. $\square$

8. THE REMAINING DIMENSION QUESTION

The example separates self-duality from the existence of a self-polar realization. Its mechanism is to combine a realizable face with a nonrealizable symmetry and enough intrinsic marking to force that symmetry as the square of every duality. Convexity of the marking operations is essential: an abstract self-dual incidence structure would not answer the realization question.

The construction does not identify the smallest possible dimension. Jensen’s 3-dimensional theorem and Theorem 1.1 place that dimension between four and ten. This leaves a concrete question.

Question 8.1. What is the smallest dimension of a realizable self-dual polytope type that admits no self-polar realization?

The present proof uses the BEK 4-dimensional face, its join with the dual type, and one additional dimension to recover both marking families. A smaller example would require a more economical way to force the same geometric obstruction, or a different obstruction.

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APPENDIX A. EXACT INCIDENCE DATA AND FINITE VERIFICATION

The construction determines a complete incidence certificate without choosing coordinates. We describe its finite rules so that the numerical assertions used in the proof can be reproduced independently. Convex realizability comes from [Proposition 4.1](#), not from these rules alone.

A.1. Recovering faces from facet sets. Let $\mathcal{H}$ be the family of facet vertex sets of a polytope, with vertex set V . For $W \subseteq V$, define its facet closure by

$$\text{cl}_{\mathcal{H}}(W) = \bigcap \{H \in \mathcal{H} \mid W \subseteq H\},$$

where an intersection over no facets is V . A vertex set is a face exactly when it equals its closure. In particular, a pair $\{u, v\}$ is an edge exactly when its closure is that pair. For a facet H , its ridges are the maximal proper members among the intersections $H \cap H'$ with other facets H' . Each such ridge has exactly two containing facets.

A.2. The generation rules. Start with the facets of Q and then of R displayed in the main text. A concrete labeling uses p_u for a vertex of the P factor and q_G for the vertex G^* of the other factor. Facet incidences of Q are

$$\mathcal{V}(A_u) = \{p_w \mid w \in \mathcal{V}(P)\} \cup \{q_G \mid u \in G\},$$

$$\mathcal{V}(B_G) = \{q_H \mid H \in \mathcal{H}(P)\} \cup \{p_u \mid u \in G\}.$$

The prism and roof incidences then follow from their displayed vertex sets.

For a cut with endpoint set W , enumerate the crossing edges uv having exactly one endpoint in W . Remove the two vertices in W . Introduce a vertex c_{uv} for each crossing edge, place it in an old facet precisely when u, v both belonged to that facet, and put all these new vertices in the cut facet. These rules retain every old facet.

For a roof addition with visible pair $\mathcal{B}$, enumerate the ridges separating a facet of $\mathcal{B}$ from a facet outside $\mathcal{B}$. Replace the two visible facets by the vertex sets $\{z_i\} \cup r$ over all these horizon ridges. Retain every other facet.

Counting membership in the resulting facet sets gives [Table 2](#). Applying the closure test verifies the middle facets, the induced BEK face, and the graph predicates used in [Lemma 5.1](#). For the BEK face specifically, take all proper traces of final facets on its ten vertices; their maximal members are exactly the 28 tetrahedra in [Table 1](#).

A.3. Companion files and symmetry enumeration. The companion directory `ancillary/` contains:

- `self_dual_counterexample.py` constructs the incidence data, verifies the finite claims, and enumerates all automorphisms and dualities;
- `self_dual_incidence.json` stores the resulting vertex–facet incidences, operation records, and symmetry maps;
- `operation_orders.py` independently checks that all 24 operation orders yield the same facet sets and agree with the JSON data.

Both programs use Python 3.9+ and only the standard library.

From that directory, the complete finite verification and regeneration are performed by

```
python3 self_dual_counterexample.py --enumerate \
    --export-json regenerated.json
python3 operation_orders.py
```

The generated JSON records the dimension supplied by the geometric construction; the checker does not infer convex realizability or dimension from an arbitrary incidence matrix.

For symmetry enumeration, form the bipartite incidence graph on 346 nodes. Initial colors distinguish vertices from facets. Preserve those colors to search for automorphisms, and exchange them in the target to search for dualities. Simultaneous refinement assigns a color according to the previous color and the multiset of neighbor colors. At a nonsingleton color class, fix one source node and branch over every target node of the corresponding class. Once all classes are singletons, verify the full adjacency relation.

Refinement signatures are invariant under every isomorphism that preserves colors, so refinement cannot discard a solution. Individualization considers every possible image of the selected node. Hence the search is exhaustive. It finds exactly two automorphisms and two dualities of the complete incidence structure. Both dualities have order four and satisfy (6.1). These group counts provide supplementary information; the exclusion of alternative dualities in the main proof uses the two marking families and does not depend on exhaustive symmetry enumeration.

The operation-order program finds identical final facet families in every order under canonical cut-vertex names, and compares them with the complete JSON. This confirms the finite incidence consequence of the commutation proof. Neither program verifies the external obstruction of Theorem 3.1 or replaces the cone arguments.

APPENDIX B. NORMALIZING SELF-POLAR REALIZATIONS

We prove the equivalence of the realization conditions stated in the introduction. All projective transformations below are defined on the whole polytope they map. Polars are taken with respect to the Euclidean inner product, and the origin lies in the interior of each polytope whose polar is used.

Proposition B.1. *If P is projectively equivalent to P° , then some polytope Q projectively equivalent to P satisfies $Q = UQ^\circ$ for an orthogonal transformation U .*

Consequently, a polytope type admits a self-polar realization if and only if it admits a realization equivalent to its polar by an isometry, an invertible linear map, an affine transformation, or a projective transformation. Isometries may include translations.

We first record the cone representation of polarity and the square of the induced correlation. This also supplies the geometric passage used in the main proof.

Lemma B.2. *If a realization of a polytope is projectively equivalent to its polar, then some combinatorial duality is induced by a projective correlation, and its square is induced by a projective automorphism.*

Proof. Let $P \subset \mathbb{R}^d$ contain the origin in its interior, let $C = \text{cone}(P \times \{1\})$, and let

$$C^\vee = \{y \in \mathbb{R}^{d+1} \mid \langle y, x \rangle \geq 0 \text{ for all } x \in C\}$$

be its positive polar cone. With $J = \text{diag}(-I_d, 1)$, we have

$$C^\vee = J \text{cone}(P^\circ \times \{1\}).$$

Suppose a projective transformation maps P° onto P . The affine denominator of this transformation is nonzero and has constant sign on the polar. Choose the overall sign of its lift G so that $G \text{cone}(P^\circ \times \{1\}) = C$. Then $L = JG^{-1}$ satisfies $L(C) = C^\vee$.

Identify covectors with vectors by the displayed pairing. The correlation sends a point represented by x to the hyperplane represented by Lx , and sends a hyperplane represented by y to the point represented by $L^{-\top}y$. Incidence is preserved because

$$\langle Lx, L^{-\top}y \rangle = \langle x, y \rangle.$$

On cone faces, the resulting duality is $F \mapsto C \cap (LF)^\perp$. Since L maps extreme rays of C to extreme rays of $C^\vee$, it exchanges vertices and facets and induces an order-reversing bijection of the whole face lattice.

The square acts on points by $T = L^{-\top}L$. Taking positive polar cones in $L(C) = C^\vee$ gives $L^{-\top}(C^\vee) = C$, so $T(C) = C$. Thus its projective action is an automorphism of P , inducing the square of the combinatorial duality. $\square$

Lemma B.3. *Let C be a pointed, full-dimensional, indecomposable polyhedral cone. If an invertible linear map T preserves C and fixes each extreme ray individually, then $T = \lambda I$ for some $\lambda > 0$.*

Proof. The extreme-ray generators are eigenvectors with positive eigenvalues and span the ambient space. Hence T is diagonalizable. If it has more than one eigenvalue, the generators partition among its eigenspaces, and C is the direct sum of the nonzero cones they generate. This contradicts indecomposability. $\square$

Lemma B.4. *Let C be a pointed, full-dimensional polyhedral cone linearly equivalent to its positive polar cone $C^\vee$. There are an invertible linear map B and an orthogonal transformation U of finite order such that*

$$U(BC) = (BC)^\vee.$$

Proof. First suppose C is indecomposable, and choose L with $L(C) = C^\vee$. Taking positive polar cones gives $L^{-\top}(C^\vee) = C$, so $T = L^{-\top}L$ preserves C . Some power T^k fixes every extreme ray. By Lemma B.3, $T^k = \lambda I$ with $\lambda > 0$. Since $\det T = 1$, we have $\lambda = 1$.

Average an inner product over the finite cyclic group generated by T , and choose coordinates orthonormal for that inner product. Under a coordinate change B , the map representing the pairing becomes $M = B^{-\top}LB^{-1}$, and its correlation square becomes $T' = BTB^{-1}$. Thus T' is orthogonal. The identities

$$M = M^\top T', \quad M^\top = (T')^\top M$$

imply $T'M = MT'$. Since $M^\top = M(T')^{-1}$, it follows that M commutes with its transpose and is normal.

Write $M = UH$ for its polar decomposition, with U orthogonal and $H = (M^\top M)^{1/2}$ positive definite. Normality implies that U and H commute. The additional coordinate change $H^{1/2}$ makes the map representing the pairing

$$H^{-1/2}MH^{-1/2} = U.$$

Hence, for $E = H^{1/2}BC$, we have $U(E) = E^\vee$. Moreover, $U^2 = M^{-\top}M = T'$, so U has finite order.

For the general case, use the canonical decomposition $C = C_1 \oplus \cdots \oplus C_r$ into indecomposable cones. To see this decomposition directly, choose one generator on each extreme ray and group generators connected by minimal linear dependencies. Every linear relation is a sum of relations with minimal support, so the spans of the groups form a direct sum. These groups give indecomposable summands, and any cone isomorphism preserves them. Choose coordinates in which their spans are mutually orthogonal; then taking positive polar cones acts on the summands separately. It also preserves indecomposability, since taking the positive polar twice returns the original cone.

The isomorphism $C \cong C^\vee$ therefore makes the multiset of linear isomorphism classes of the summands invariant under taking positive polars. Each class equal to its own polar class is normalized by the preceding argument. The remaining classes occur in pairs with equal multiplicities. A pair can be realized as $D \oplus D^\vee$ in two orthogonal copies of the same Euclidean space. The orthogonal map $(x, y) \mapsto (y, x)$ sends this cone onto $D^\vee \oplus D$, its positive polar cone, and has order two. Combining the maps on all summands gives the required finite-order orthogonal map on a cone linearly equivalent to C . $\square$

Proof of Proposition B.1. Let $C = \text{cone}(P \times \{1\})$. The construction in Lemma B.2 gives a linear equivalence between C and $C^\vee$. By Lemma B.4, a cone E linearly equivalent to C satisfies $U(E) = E^\vee$ for a finite-order orthogonal transformation U . Taking positive polar cones gives $U(E^\vee) = E$.

We need a point in $\text{int } E \cap \text{int } E^\vee$. Choose a covector ℓ strictly positive on $E \setminus \{0\}$ and minimize the Euclidean norm on the compact section $\{x \in E \mid \ell(x) = 1\}$. Its minimizer p satisfies $\langle p, x \rangle \geq \|p\|^2$ for every point x of the section, by differentiation along the segment from p to x . Consequently,

$$\langle p, z \rangle \geq \|p\|^2 \ell(z) > 0 \quad (0 \neq z \in E).$$

Thus $p \in \text{int } E^\vee$. Perturbing p slightly into $\text{int } E$ gives a point v in the common interior.

The common interior is invariant under U . If $U^N = I$, its average

$$w_0 = \frac{1}{N} \sum_{j=0}^{N-1} U^j v$$

still belongs to the common interior and is fixed by U . In particular, $w_0 \neq 0$. Normalize it to a unit vector w and take the bounded section

$$Q = \{q \in w^\perp \mid w + q \in E\}.$$

It is full-dimensional in $w^\perp$ and contains its origin in its interior. Its polarity is taken within this Euclidean space. The corresponding section of $E^\vee$ is

$$\{y \in w^\perp \mid w + y \in E^\vee\} = -Q^\circ,$$

since $\langle w + y, w + q \rangle = 1 + \langle y, q \rangle$. As U fixes w , its restriction U_0 to $w^\perp$ satisfies $U_0 Q = -Q^\circ$. Therefore $Q = -U_0^{-1} Q^\circ$, as required.

An invertible linear change of a homogenized cone followed by a bounded section is a projective transformation of the original polytope. Its denominator is strictly positive on that polytope. Thus Q is projectively equivalent to P .

Finally, orthogonal equivalence implies linear, affine, isometric, and projective equivalence. Conversely, each of these equivalences is a projective transformation, so the proved normalization applies. This establishes the equivalence of all the existence conditions, including isometries with translations. $\square$